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PROJECT TITLE:
EcoEducation – SDG 7

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1.0 Introduction

EcoEducation is a mobile learning application developed to support Sustainable Development Goal (SDG 7: Affordable and Clean Energy). This project focuses on educating students and young people about renewable energy sources such as solar energy, wind energy, hydropower, and biomass energy.

In Malaysia, renewable energy is becoming more important to reduce environmental pollution and carbon emissions. However, awareness and understanding of renewable energy among students is still low. Therefore, EcoEducation is designed as an interactive learning platform that helps users gain knowledge through structured lessons, quizzes, and progress tracking.

This application also provides an engaging learning experience by allowing users to explore renewable energy topics, answer quizzes, and monitor their progress, making the learning process more enjoyable and effective.

2.0 Problem Statement

2.1 Problem Statement

Many students and young people in Malaysia have limited awareness and knowledge about renewable energy sources such as solar, wind, hydropower, and biomass. Most learning materials about renewable energy are presented in a theoretical and text-heavy way, which makes it difficult for beginners to understand.

In addition, there are limited mobile learning applications that focus specifically on SDG 7 and provide interactive features such as quizzes and learning progress tracking. This lack of accessible and engaging learning tools reduces student interest in renewable energy education. Without proper understanding, society may be slower to support clean energy adoption, which can affect Malaysia's long-term sustainability goals and progress in reducing pollution and carbon emissions.

2.2 Problem Statement (Points)

- Many students have low awareness of renewable energy topics.
- Renewable energy learning is often too theoretical and difficult to understand.
- Existing learning materials are boring and not interactive.
- Limited mobile applications focus on SDG 7 renewable energy education.
- Students lack a platform to test their knowledge through quizzes.
- No structured learning system that tracks user progress.
- Low renewable energy knowledge may reduce support for clean energy adoption.

3.0 Proposed Solution

3.1 Proposed Solution

To solve the issue of low renewable energy awareness among students and young people in Malaysia, EcoEducation was developed as a mobile learning application that focuses on SDG 7: Affordable and Clean Energy. The app provides structured learning content about renewable energy sources such as solar energy, wind energy, hydropower, and biomass energy.

EcoEducation delivers learning materials in a simple and beginner-friendly format through chapter-based lessons. After completing each chapter, users can answer quizzes to test their understanding. The app also provides progress tracking features that allow users to monitor their learning achievements. In addition, EcoEducation includes a quiz game mode to make learning more interactive and enjoyable.

Through this solution, users are encouraged to explore renewable energy topics and build awareness about clean energy adoption, supporting Malaysia's sustainability goals.

3.2 Proposed Solution

- Develop a mobile learning app called EcoEducation focused on SDG 7.
- Provide renewable energy lessons in a chapter-based learning system.
- Cover four main renewable energy topics: Solar, Wind, Hydropower, and Biomass.
- Include quizzes after each chapter to test user understanding.
- Provide a progress tracking screen to monitor learning completion.
- Implement a shared ViewModel state to update progress across different screens.
- Add an interactive EcoQuiz mode with difficulty levels to increase engagement.
- Encourage students to learn renewable energy in a simple, fun, and user-friendly way.

4.0 Objectives

4.1 General Objective

To develop a mobile learning application that increases awareness and understanding of renewable energy to support SDG 7: Affordable and Clean Energy.

4.2 Specific Objectives

- To provide educational content about renewable energy (Solar, Wind, Hydropower, Biomass).
- To create an interactive learning system with quizzes to test user understanding.
- To allow users to track their learning progress using progress indicators.
- To encourage students to learn renewable energy through a simple and user-friendly interface.
- To support Malaysia's sustainability efforts by improving renewable energy awareness.

5.0 Conclusion

In conclusion, EcoEducation is a mobile learning application designed to address the lack of renewable energy knowledge among students and young people in Malaysia. The project supports SDG 7 by providing structured renewable energy lessons, quizzes, and progress tracking features.

EcoEducation helps users learn important concepts related to clean energy in an engaging way and encourages them to understand the importance of renewable energy adoption. With this application, students can improve their knowledge and become more aware of sustainable energy solutions, which contributes to a cleaner and greener future.